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for managing natural resources and a better life for all”

Stakeholder engagement in agro-climate service planning

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Abstract

The impacts of weather, climate variability and climate change on agricultural production underline the increasing importance of actionable agro-climatic services for improving the resilience of small-holder farmers. While, in general, the provision of climate services has increased, there is often a mismatch between service providers and users about what constitutes useful information. Transitioning from supply-driven provision of climate and agricultural information to demand-driven agro-climate services (ACS) at scale cannot be accomplished in a top-down manner but requires the engagement of diverse stakeholders in all phases of ACS, including the designing and budgeting phases. However, such pre-stage transitions often feature uncertainty, scattered knowledge, conflicts of different views, and the challenge of bringing stakeholders to the same table.

Our study offers a transparent and systematic method to address critical challenges in the early stages of ACS development by engaging stakeholders in generating and translating system knowledge for decision-making. Our findings reveal that considering multiple stakeholder attributes — such as availability, experience, gender, expertise, cost and benefit profile, interest, influence, relevance, and attitude — is critical for effective stakeholder engagement. Combining these attributes with stakeholders' system knowledge and understanding of the decision-making process, in a case study in Dien Bien, Vietnam, we identified when, where, and how stakeholders could engage in the socio-economic development planning process. For example, stakeholders with extensive experience and relevant expertise should focus on producing evidence about the needs and impacts of ACS and include it in formal reports to encourage political interest. Influential stakeholders with an interest in and relevance to ACS should be involved in key decision-making moments within the policy cycle. If interest among influential stakeholders is limited, a preparatory process that provides credible, legitimate, and relevant evidence is recommended. Throughout the stakeholder engagement process, we emphasised the need for a coordinating body that is viewed as relevant to decision-making yet has minimal conflicts of interest.

Our proposed engagement approach guides the development of ACS that is responsive to local needs for climate change response. Our methods can be applied to support other complex decision-making processes.

Keywords: Complexity, decision analysis, stakeholder attributes, uncertainty, upscaling